Cloud Computing Case Study Project Plan

Week	Tasks	Deliverables
Week 1	• Find primary sources, relevant information, extract architecture, cost and outcome data	One page summary of • sources found • Case study salient points
Week 2	Case analysis: Detailed study architecture diagram, problem, solution	Submit 10-page report on study, exercise plan, roles of team members
Week 3	• build the core of the exercise • run experiments and collect measurements (latency, scaling behavior, cost) • Test system and update with final results	Add-ons to report with results
Week 4	Report preparation and submission	
Week 5	Project VIVA	

Case Study Topics:

Scalability & High-Traffic Events

1. **Airbnb** → Service-oriented architecture, holiday traffic spikes
2. **IRCTC** — Cloud scaling for railway ticket booking (handles massive simultaneous demand, e.g., Tatkal booking spikes)
3. **Hotstar (Disney+ Hotstar)** — Cloud scaling for record-breaking cricket streaming concurrency (IPL/World Cup)
4. **Flipkart** — Big Billion Days traffic scaling (hybrid cloud + own infra evolution)
5. **Myntra** — Cloud-based personalization and flash-sale scaling
6. **Zomato** — Multi-region cloud architecture for food delivery at scale

Migration & Transformation Stories

1. **Netflix** → AWS migration, Chaos Monkey/resilience
2. **Capital One** → Full exit from data centers to AWS (banking/regulated industry)
3. **GE (General Electric)** → Predix industrial IoT cloud platform
4. **Philips** → HealthSuite digital platform on cloud
5. **Titan Industries** and **Tata Motors** — Put digital business and properties on AWS, using IoT and telemetry

"Repatriation" / Moving Away from Cloud

1. **Dropbox** → Built "Magic Pocket" to leave AWS S3
2. **Basecamp (37signals)** → Publicly left the cloud, wrote about savings (2022–23)
3. **Meesho** — Cost-efficient cloud scaling for social commerce in tier 2/3 India

Startups & AWS Case Studies

1. **Shaadi.com** — Uses AI and machine learning from AWS to analyze 67 billion event records to make 4 million matches
2. **Practo** — Connects 60 million patients to 200,000 doctors from 10,000 hospitals using AWS
3. **Freshworks** — SaaS company built and scaled on cloud from India for global markets

Government & Public Digital Infrastructure (India-specific strength area)

1. **Aadhaar (UIDAI)** — World's largest biometric ID system, cloud/data-center architecture at national scale
2. **CoWIN** — Cloud platform built for India's COVID-19 vaccination scheduling at national scale
3. **DigiLocker** — Cloud-based document storage/verification, part of India Stack

Case Study based exercises

Theme (example cases)	Exercise
Scalability (Hotstar, IRCTC, Flipkart, Zomato)	Deploy a small web app behind a load balancer with autoscaling, run a load test (e.g., k6 or Locust), and plot response time against instance count.
Migration (Tata Motors, Capital One, Netflix)	Lift-and-shift a simple app from a local Docker container to a cloud VM, then refactor part of it to a managed service. Document effort and cost differences.
Cost and repatriation (Dropbox, Basecamp, Meesho)	Use the provider pricing calculators to model the case's workload on two providers, then compare against an on-premises estimate.
Serverless and startups (Shaadi.com, beauty-retail streaming case)	Build a small serverless function with an API trigger and measure cold start and cost per 1,000 requests.
Government and public platforms (Aadhaar, UPI, CoWIN, DigiLocker)	Design a multi-region, high-availability architecture for a national-scale service, then build a scaled-down version of one component.